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**B.Tech. Degree VI Semester Special Supplementary Examination
November 2014**

**ME 603 MACHINE DESIGN I
(2006 Scheme)**

Time : 3 Hours

Maximum Marks : 100

**PART A
(Answer ALL questions)**

(8 × 5 = 40)

- I. (a) Define endurance limit. Explain the factors affecting endurance limit.
- (b) Explain the following.
 - (i) Factor of safety.
 - (ii) Methods to reduce stress concentration.
- (c) What is bolts of uniform strength? Explain with examples, the practical applications of such bolts.
- (d) Discuss the various types of shaft couplings, mentioning the use of each type.
- (e) Explain how the efficiency of a riveted joint is determined.
- (f) What are the important factors to be considered in the selection of material for the spring wire?
- (g) Show that the normal stress in case of an annular fillet weld subjected to bending is given by

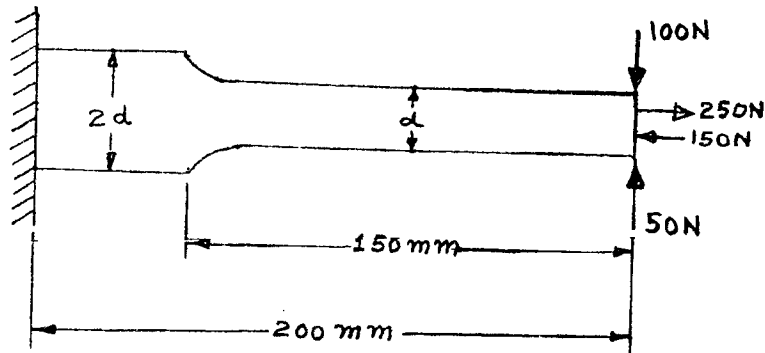
$$\sigma = \frac{5.66 M_b}{\pi s d^2}$$
 where M_b = bending moment, S = weld size, d = diameter of cylindrical element welded to flat surface.
- (h) Explain the following terms.
 - (i) Equivalent torsional moment
 - (ii) Equivalent bending moment.

PART B

(4 × 15 = 60)

- II. A steel cantilever rod is subjected to an axial load which varies from 150 N in compression to 250 N in tension and also a transverse load at its free end which varies from 50 N up and 100 N down as shown in figure. Determine the diameter d , assuming the following values: Factor of safety = 2, $\sigma_{field} = 330 \text{ N/mm}^2$, $\sigma_{endurance} = 300 \text{ N/mm}^2$, $K_{bending} = 1.4$, $K_{axial} = 1.6$, q (notch sensitivity) = 0.9, A (correction factor for type of loading) = 1, B (size correction factor) = 0.85, C (surface correction factor) = 0.9. (15)

(P.T.O.)



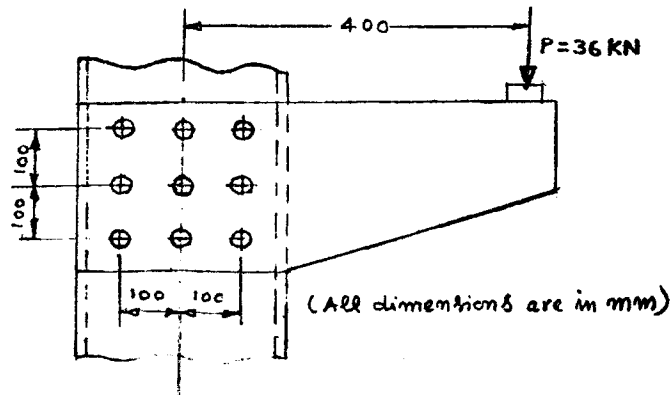
OR

- III. A simply supported beam has a concentrated load at the centre which fluctuates from a value of 10 N to 40 N. The span of beam is 400 mm and its cross section is circular. Taking for the beam material an ultimate stress of 700 MPa, any field stress of 500 MPa, endurance limit of 330 MPa for reverse bending and factor of safety = 1.3, calculate the maximum value of diameter, d , using Goodman's relation and Soderberg relation. Take a size correction factor of 0.85 and surface finish correction factor of 0.9. (15)

- IV. A vertical two start square threaded screw of 100 mm mean diameter and 20 mm pitch supports a vertical load of 18 kN. The nut of the screw is fitted in the hub of a gear wheel having 72 teeth which meshes with a pinion of 18 teeth. The mechanical efficiency of the pinion and gear wheel drive is 85 percent. The axial thrust acting on the screw is taken by a collar bearing 240 mm outside diameter and 90 mm inside diameter. Assuming uniform pressure conditions, find minimum diameter of pinion shaft and height of nut if coefficient of friction for the vertical screw and nut is 0.15 and that for the collar bearing is 0.20. The permissible shear stress in the shaft material is 50 MPa and allowable bearing pressure 1.3 MPa. (15)

OR

- V. Design and draw a cotter joint for connecting two rods which are subjected to a tensile force of 32 kN. The allowable stress in tension, shear and crushing may be assumed as 55 MPa, 45 MPa and 100 MPa respectively. (15)
- VI. A bracket is attached to a steel channel by means of nine identical rivets as shown in figure. Determine the diameter of rivets, if the permissible shear stress is 55 N/mm². (15)



OR

(Con...3)